

# AJAY KUMAR GOLLA

+1 408-690-5137 | ajaykumargolla229@gmail.com | LinkedIn | San Jose, CA

## SUMMARY

Software Engineer with industry and research experience building distributed backend systems, cloud-native infrastructure, and production AI platforms using Python, Go, Java, Kubernetes, AWS, and GCP.

## EXPERIENCE

### Research Assistant — San Jose State University

Jan 2025 - May 2026

- Designed cloud-native backend services using Go, Kubernetes, Docker, and AWS, improving throughput by 30% through asynchronous processing, distributed service orchestration, and optimized resource utilization.
- Built Dockerized REST services and distributed data pipelines, reducing execution latency by 25% through workload parallelization, concurrency optimization, and efficient backend service design.
- Benchmarked and optimized distributed systems under production-like workloads, identifying performance bottlenecks to improve scalability, reliability, and observability for cloud-native AI infrastructure.

### Software Engineer — Rivach LLP

June 2023 – July 2024

- Architected and deployed stateless Spring Boot microservices with well-defined REST APIs, supporting 5K+ users while enabling horizontal scalability, fault isolation, and reliable distributed service communication.
- Reduced API latency by 30% under production load by optimizing SQL query execution plans, implementing Redis caching, and restructuring data access patterns to improve throughput across high-traffic backend services.
- Engineered containerized CI/CD pipelines using Docker, Jenkins, and AWS, increasing deployment frequency by 40% through automated build, testing, deployment, and rollback workflows, improving release reliability and operational efficiency.

## PROJECTS

### Oprina – Production Multi-Agent AI Platform | *Python, FastAPI, Gemini 2.0, Vertex AI, GCP Cloud Run, Supabase, Docker*

- Built and deployed a production multi-agent AI platform on GCP Cloud Run, orchestrating specialized agents for Gmail, Calendar, and conversational workflows with low-latency tool execution and real-time user interactions.
- Designed a centralized orchestration layer for multi-step agent execution, implementing session persistence, tool invocation workflows, and retrieval-backed memory to maintain reliable long-running conversations.
- Improved platform reliability through monitoring, failure recovery, and optimized context management, enabling scalable deployment of long-running AI agent workflows on GCP Cloud Run.

### RepoPilot AI – Autonomous Coding Debugging Platform | *Python, LangGraph, FastAPI, PostgreSQL, Docker, MCP, LangFuse, FAISS*

- Developed a multi-agent software engineering platform using LangGraph and MCP, orchestrating autonomous workflows across GitHub, Jira, Slack, and PostgreSQL through reusable tool integrations.
- Built hybrid retrieval pipelines using FAISS and Tree-sitter to provide semantic code search and context-aware reasoning, improving retrieval quality for complex engineering tasks.
- Designed LangFuse evaluation pipelines to measure retrieval quality, tool-call reliability, latency, and agent performance, enabling systematic experimentation and iterative optimization through failure analysis.

### Reddit Distributed Messaging Platform | *Node.js, Kafka, Redis, MySQL, MongoDB, AWS EC2*

- Designed an event-driven distributed messaging platform using Kafka producers/consumers and Redis caching, supporting reliable asynchronous communication across distributed services.
- Implemented fault-tolerant messaging workflows with retry handling, asynchronous processing, and optimized database access, improving throughput by 70% while reducing end-to-end latency by 45%.
- Developed scalable REST APIs supporting 5K+ concurrent users across a multi-node AWS deployment, ensuring reliable message delivery through fault-tolerant event-driven workflows.

### MindMesh – Distributed Knowledge Graph Platform | *Kotlin, Elixir, PostgreSQL, MongoDB, Redis, Docker*

- Architected a distributed microservices platform with reactive backend services and an API gateway, supporting 1K+ concurrent users with fault-isolated service communication.
- Designed a hybrid storage architecture using PostgreSQL, MongoDB, and Redis to optimize transactional consistency, hierarchical data access, and high-throughput caching.
- Improved system scalability through load balancing, circuit breakers, and distributed caching, achieving 99.9% service availability under concurrent workloads.

## TECHNICAL SKILLS

**Languages:** Python, Go, Java, C++, JavaScript, TypeScript, SQL

**Backend & Frameworks:** Spring Boot, FastAPI, Flask, Node.js, Express.js, REST APIs, GraphQL, gRPC, Microservices

**Distributed Systems:** Kafka, Redis, Event-Driven Architecture, Asynchronous Processing, Distributed Systems, Fault Tolerance, Load Balancing  
**AI & Search:** LangGraph, LangChain, MCP, RAG, FAISS, Neo4j, Vector Search, Hybrid Retrieval, Semantic Search, Information Retrieval, Prompt Engineering, LLM Evaluation

**Cloud & DevOps:** AWS (EC2, S3, RDS, Lambda), GCP (Vertex AI, Cloud Run), Kubernetes, Docker, Jenkins, Git, GitHub Actions, CI/CD, Linux

**Databases & Messaging:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j, Supabase

**Observability & Testing:** LangFuse, Structured Logging, Performance Profiling, Jest, React Testing Library

## EDUCATION

**San Jose State University** - M.S. Computer Software Engineering

Aug 2024 - May 2026

Relevant Coursework: Distributed Systems, Advanced Algorithms, Cloud Computing, Machine Learning Systems, Deep Learning, Data Mining